

Final Test

Question 1

Which of the following is NOT a function of investment banks?

☐

 Providing research for clients

☐

 Accepting deposits from the general public

☐

 Underwriting securities

☐

 Assisting with IPO issuance

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QUESTIONS

X	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X
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X	X	X	X

Question 2

What is the main purpose of financial covenants in lending agreements?

☐

 To ensure that the company operates within certain rules set by the lenders

☐

 To encourage the company to take on more debt

☐

 To allow the company to borrow as much money as possible

☐

 To protect the company from having to repay the loans

Question 3

Why do bonds issued by corporations generally have higher yields than bonds issued by the federal government?

- ☐ Corporate bonds have credit risk while government bonds are considered risk-free
- ☐ Government bonds are less liquid than corporate bonds
- ☐ Government bonds are more volatile than corporate bonds
- ☐ Corporate bonds have longer maturities than government bonds

Question 4

Assume that you deposit \$15,000 at a local bank. The bank offers 4% interest rate. You keep the money deposited for 5 years. How much money will you have in your deposit account after 5 years, assuming (a) compound interest, and (b) simple interest. (Round to the nearest integer.)

- ☐ \$18,250; \$15,600
- ☐ \$18,000; \$18,250
- ☐ \$18,250; \$18,000
- ☐ 18,000,15,600

Question 5

Company LMN made an investment of \$10,000. This investment will pay the company the following cash flows: Year 1 \$0, Year 2 \$3,000, Year 3, \$0, and Year 4: \$12,500. (a) What is the present value of the future cash flows, assuming the suitable discount rate is 5%? (b) What is the Net Value of the investment of the company (Note: LMN invested \$10,000)? (Round to nearest integer.)

- ☐ \$13,520,0

- ☐ \$10,000; \$3,005
- ☐ \$13,005; \$0
- ☐ \$13,005; \$3,005

Question 6

Which of the following best describes the relationship between risk and potential return for small-cap stocks compared to large-cap stocks?

- ☐ Small-cap stocks generally have higher risk and higher potential returns
- ☐ There is no difference in risk or potential return between small-cap and large-cap stocks
- ☐ Small-cap stocks always have lower risk and higher returns
- ☐ Small-cap stocks have lower risk but also lower potential returns

Question 7

What is the primary difference between the coefficient of variation and the Sharpe ratio?

- ☐ The coefficient of variation uses standard deviation while the Sharpe ratio uses variance
- ☐ The Sharpe ratio uses excess return over standard deviation
- ☐ The coefficient of variation is always larger than the Sharpe ratio
- ☐ The Sharpe ratio is only applicable to stocks while the coefficient of variation can be used for any asset

Question 8

If a stock's daily returns have an excess kurtosis of 5, what can we conclude about its distribution?

- ☐ The distribution is a non-Gaussian distribution
- ☐ The distribution is likely negatively skewed
- ☐ The distribution is perfectly normal
- ☐ The distribution has thinner tails than a normal distribution

Question 9

MNQ shares are currently trading for \$22/share. We just bought 100 shares, and then 1 year later we sell them for \$24/share. During the period MNQ paid dividends in the amount of \$1/share. What is our rate of return on this investment? (Assume that there are no transaction fees and commissions.)

- ☐ 13.64%

- ☐ 10.05%
- ☐ 8.70%
- ☐ 9.09%

Question 10

Current stock price \$45 per share We are given the following information about company ABC:

- Cost of equity 8.5%
- Last dividend paid \$2 per share
- Growth rate stage 1 (fast growth): 8%
- Growth rate stage 3 (stable growth): 3.5%
- Stage 2 (linearly declining growth rate) is expected to last 8 years (i.e., $H=8$) (a) Using the H model what is the value of ABC shares? (b) Are they currently over/under or correctly valued?

- ☐ \$48.60, overvalued
- ☐ \$48.60, undervalued
- ☐ \$55.80, overvalued
- ☐ \$55.80, undervalued

Question 11

Which of the following is NOT a benefit of ETFs?

- ☐ Low fees
- ☐ Trading flexibility
- ☐ Guaranteed outperformance of the market
- ☐ Professional management

Question 12

What is the main advantage of using the Sharpe ratio over the Coefficient of Variation?

- ☐ The Sharpe ratio accounts for the risk-free rate
- ☐ The Sharpe ratio is always positive
- ☐ The Sharpe ratio is easier to calculate
- ☐ The Sharpe ratio works better for non-normal distributions

Question 13

How might you create a hedging strategy using ETFs to protect against potential market downturns?

- ☐ Combine long positions in broad market ETFs with short positions in volatile sector ETFs or inverse ETFs
- ☐ Increase leverage by borrowing to buy more of the same ETFs
- ☐ Invest all assets in a single sector ETF
- ☐ Sell all ETF holdings and hold only cash

Question 14

Suppose that we are considering investing in 4 possible funds. We are provided with the following annualized return and volatility for each of them over the last 10 year period:

- Fund A 15% return 22% volatility
 - Fund B 12% return, 20% volatility
 - Fund C 10% return, 8% volatility
 - Fund D 25% return 30% volatility
- Assume the risk-free rate is 2%. Which fund should we choose based on the Sharpe ratio?

- ☐ Fund A
- ☐ Fund D
- ☐ Fund B
- ☐ Fund C

Question 15

DEF is an ETF that has assets of \$100 million and liabilities of \$70 million. The ETF has 10,000 shares

outstanding. What is DEF's NAV?

- ☐ \$10,000
- ☐ Cannot be determined
- ☐ \$7,000
- ☐ \$3,000

Question 16

How does the payoff of an option differ from that of stocks or futures?

- ☐ Options have an exponential payoff
- ☐ Options have a non-linear payoff
- ☐ Options have a logarithmic payoff
- ☐ Options have a linear payoff

Question 17

Which of the following is NOT a fundamental type of derivative?

- ☐ Futures
- ☐ Bonds
- ☐ Options
- ☐ Forwards

Question 18

Design a options strategy that would benefit from a significant move in the underlying asset's price in either

direction, but with limited risk.

- ☐ Buy a call option with a high strike and sell a put option with a low strike
- ☐ Sell a call option with a high strike and buy a put option with a low strike
- ☐ Buy a call option and a put option with the same strike price and expiration (long straddle)
- ☐ Sell a call option and a put option with the same strike price and expiration (short straddle)

Question 19

How does the time value impact the value of (a) ATM Call option, (b) ATM put option. The answer should specify if the time value is positive or negative for both types of options.

- ☐ Call Positive, Put negative
- ☐ Call Negative, Put Negative
- ☐ Call Negative, Put positive
- ☐ Call Positive Put positive

Question 20

You bought a call option on FGH with a strike price of \$40 that expires in 6 months. The current price of the call is \$4 and the current share price of FGH is \$43 per share.
(a) What is the intrinsic value of the call?
(b) What is the time value of the call?

- ☐ Intrinsic value 0, time value potentially unlimited
- ☐ Intrinsic value = 3, time value =1

- ☐ Intrinsic value 0, time value =4
- ☐ Intrinsic value=1, time value =3

Question 21

What potential impact could widespread mortgage refinancing have on MBS investors?

- ☐ It would always result in higher returns for all investors
- ☐ It would automatically trigger defaults on all MBS tranches
- ☐ It could lead to reinvestment risk as principal is returned earlier than expected
- ☐ It would have no effect on MBS cash flows

Question 22

What is the primary purpose of the Dodd-Frank Act?

- ☐ To promote international banking
- ☐ To simplify financial products
- ☐ To increase bank profits
- ☐ To prevent another financial crisis

Question 23

How might one critically assess the argument that increased regulation is necessary to prevent financial crises?

- ☐ By disregarding the historical context of past financial crises
- ☐ By weighing the potential benefits against the risks of unintended consequences
- ☐ By assuming that all regulation is inherently detrimental
- ☐ By focusing solely on the costs of implementing new regulations

Question 24

Assume that a bank has made a loan of \$300 mln to firm XYZ with no collateral required. The bank estimates that in case firm XYZ defaults, the RR (recovery rate) is 0.60.

(a) What is the LDG (loss given default) dollar amount faced by the bank?

(b) How would the LDG dollar amount change if the bank had asked for \$150 mln of collateral in order to provide the loan, assuming that the collateral can be easily sold for that value.

- ☐ \$300 mln; \$20 mln
- ☐ \$120 mln; \$60 mln
- ☐ \$180 mln; \$120 mln
- ☐ \$150 mln; \$10 mln

Question 25

Assume that the risk-free interest rate (RFR) is currently 1.5%. Suppose that we are interested in a bond ZZZ whose yield (BY) is currently 5.7%. The recovery rate (RR) for this type of bonds is 75%. Compute the Probability of Default (PD) for bond ZZZ.

- ☐ 22.80%

- ☐ 5.60%
- ☐ 16.80%
- ☐ 21.80%

Question 26

Which factor is least relevant when deciding whether to rent or buy a home?

- ☐ The color of the house
- ☐ The potential for property value appreciation
- ☐ The local housing market trends
- ☐ The individual's long-term financial goals

Question 27

How does selling equity in a real estate development company affect profit distribution?

- ☐ It increases profits for existing shareholders
- ☐ It means sharing profits with additional parties
- ☐ It has no effect on profit distribution
- ☐ It eliminates the need for profit distribution

Question 28

How might a microfinance institution benefit from providing Housing Microfinance (HMF)?

- ☐ Increased government subsidies

- ☐ Elimination of credit risk
- ☐ Diversification of its loan portfolio
- ☐ Lower interest rates on loans

Question 29

Assume that a company has Assets of JPY 120 bln and Liabilities of of JPY 100 bln.

- (a) What is the equity multiplier? (Round to 1st decimal.)
(b) Suppose that now the company has spent JPY 20 bln to pay down some debt it owed to a bank. What is the equity multiplier now?

- ☐ 5.6x; 4.2x
- ☐ 1.2x; 1.3x
- ☐ 5.5x; 4.7x
- ☐ 6.0x; 5.0x

Question 30

Assume that Bank DDF has EUR 20,000,000 in cash, securities worth EUR 200,000,000, loans totaling EUR 300,000,000 and other assets worth EUR 180,000,000. Then bank has also deposits for EUR 210,000,000 and other borrowings for EUR 60,000,000 and, finally, DDF has EUR 20,000,000 in capital reserves. What is Bank DDF's equity?

- ☐ EUR 450 mln
- ☐ EUR 430 mln
- ☐ EUR 850 mln
- ☐ EUR 410 mln

Question 31

Which of the following best describes the impact of increased correlations between assets during market stress?

- ☐ It reduces the benefits of diversification
- ☐ It only affects bond portfolios not equity portfolios
- ☐ It increases the benefits of diversification
- ☐ It has no effect on portfolio risk

Question 32

Analyze the potential consequences of assuming constant volatility in option pricing models.

- ☐ It always leads to overpricing of options
- ☐ It has no impact on option pricing accuracy

- ☐ It may lead to mispricing of options, especially for longer-dated or out-of-the-money options
- ☐ It always results in more accurate option prices

Question 33

Evaluate the effectiveness of using copulas instead of simple correlation in modeling the dependence structure of a multi-asset portfolio.

- ☐ The choice between copulas and correlation has no impact on portfolio risk assessment
- ☐ Copulas and simple correlation yield identical results in all scenarios
- ☐ Copulas are always less effective than simple correlation
- ☐ Copulas can capture non-linear dependencies and tail risks more effectively than simple correlation

Question 34

A bond with a maturity of 5 years, a coupon rate of 4%, and a yield to maturity of 3.5% has a modified duration of 4.0. If the yield decreases by 0.3% what is the approximate change in bond price as a percentage?

- ☐ -1.20%
- ☐ -0.01%
- ☐ 1.20%
- ☐ 0.01%

Question 35

For a bond that has a yield to maturity of 7% and whose \$-duration is -350, what is the absolute gain (loss) when the yield to maturity increases by 0.3%

- ☐ \$105 gain
- ☐ \$1.05 loss
- ☐ \$1.05 gain
- ☐ \$105 loss

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